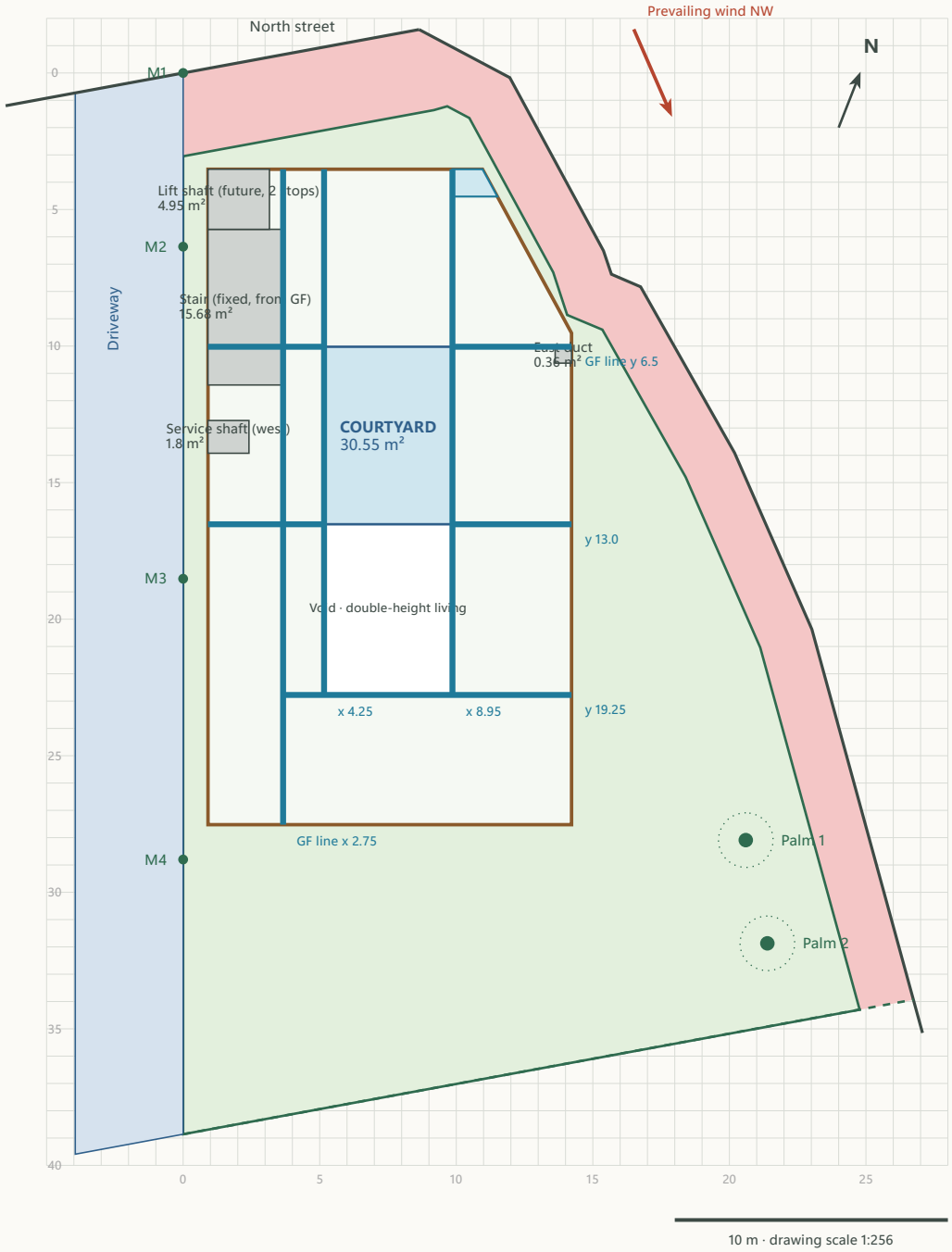


Taif / Palm House

First floor · starting point · on ground floor V4.0

16 Oct 2026 · base.typ + design_ff.typ + data/rooms_ff.csv · constraints in docs/FIRST_FLOOR_BRIEF.md

First floor, not designed yet. Drawn from the ground floor V4.0: the outline (13.311 × 24.0 m on wall centre lines), the **voids** with no floor (the courtyard, open to the sky, and the double-height living), and the **fixed vertical elements** in grey: the stair (2.75 × 5.7), the future lift shaft (two stops, ground and first), the west service shaft and the east duct; the air tower (blue) rises to the roof. Blue lines are ground-floor wall lines that can carry first-floor walls. Storey height assumption 3.6 m floor to floor. Add rooms to data/rooms_ff.csv (same CSV coordinates as the ground floor); first-floor bathrooms over or beside the west shaft (north part of the west wash block) or the east duct. 0 first-floor rooms so far.

✓ **All checks passed**

Areas

Envelope	**667.87 m²**
Building limit (envelope unless set)	**309.68 m²**
Other open voids	**29.38 m²**
Courtyard (open to sky)	**30.55 m²**
Rooms placed (4)	**22.79 m²**
Limit minus courtyard and voids	**249.76 m²**
Air channel + duct	**1.37 m²**
Not yet assigned	**226.97 m²**

Orientation

True compass vs the M line	**21.4°**
NW wind vs the M line	**23.6°**

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

Envelope corners

Corner	x	y
V1	0	38.86
V2	0	3.05
V3	9.18	1.36
V4	9.67	1.22
V5	10.48	1.65
V6	13.55	7.3
V7	14.06	8.86
V8	15.35	9.4
V9	18.4	14.8
V10	21.13	21.04
V11	24.77	34.3

Rooms

Name	Frame	w × d (m)	Area
Stair (fixed, from GF)	M	2.75 × 5.7	15.68 m²
Lift shaft (future, 2 stops)	M	2.25 × 2.2	4.95 m²
Service shaft (west)	M	1.5 × 1.200000000000001	1.8 m²
East duct	M	0.5999999999999996 × 0.5999999999999996	0.36 m²